Weekly Lesson Plan – Week 3 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon | Week of: Aug 24, 2026 | Course: Engineering Design & Problem Solving (InSPIRESS)

TEKS / ELPS:

TEKS: §127.785 present and defend a design solution to a review panel; §127.785 evaluate design solutions; give and receive technical critique; §127.785 use digital design tools; set up the CAD environment; §127.785 professional ethics and responsibility in engineering practice; §127.785 ethical decision making; justify and defend a professional judgment
ELPS: c.1A, c.2D, c.3C, c.4G, c.5B

Aim of the Lesson / Objective: *Statement of what you want students to do (skill) and/or know (knowledge). Students will…*

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
Students will present their product to a review panel and defend it under questioning, and evaluate peer designs against a shared rubric.	Students will finish presenting and defending their designs, then identify what the strongest design solutions had in common.	Students will analyze an engineering failure in writing and set up and sign in to a working Fusion 360 account.	Students will explain the engineer's first obligation to public safety and analyze a real engineering failure to identify the decision point.	Students will make and defend an ethical decision about a flaw in their own product, giving reasons and naming the cost of their choice.

Employed Language Domain(s) Objective: *How students achieve the objective – Listening, Speaking, Reading, Writing. Students will…*

Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members' ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings.

Assessment of Content Objective: *Instrument used to determine mastery. Ex: Exit Ticket.*

Students will be assessed by written, group, or oral responses throughout the class period.

Set (Bell Work): *A motivating action that engages students in the objective – question, fact, activity. Essential Question:*

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
What is the one question you are most afraid the panel will ask you?	Which product you saw yesterday solved the most real problem, and why?	If you knew a product you designed had a problem, but fixing it meant missing the deadline, what would you do?	You find a mistake in your team's work on the day it is due. Nobody else noticed. What do you do, and why?	Is it ever acceptable to sell something you know has a small flaw? Explain.

Methodology / Steps: *What it takes to teach the objective so 100% can pass the assessment. Teacher will… Teams will… Individuals will…*

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
Step 1: Bell ringer / essential question. Step 2: Bell-work: the question you're most afraid of (5 min) Step 3: Order of business + judging rules (5 min) Step 4: Design Reviews: 3-min pitch + 2-min defense per firm (30 min) Step 5: Turn in peer scorecards (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: which product solved the most real problem? (5 min) Step 3: Remaining firms present + defend (28 min) Step 4: Class debrief: what every strong pitch had (7 min) Step 5: Best Engineering Judgment + exit (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: fix it or hit the deadline? (6 min) Step 3: Video + 4 written response questions (24 min) Step 4: Fusion 360 account setup (rolling, row by row) (10 min) Step 5: Confirm every account signs in (5 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work (ethical): you find a mistake the day it's due (6 min) Step 3: The code: who an engineer protects first (10 min) Step 4: Firms analyze an assigned real case (five questions) (20 min) Step 5: 60-second report out per firm (9 min)	Step 1: Bell ringer / essential question. Step 2: Bell-work: is a small flaw ever acceptable? (6 min) Step 3: The situation + the three options + worked example (8 min) Step 4: Firms decide, write 3 reasons, and draft the customer statement (18 min) Step 5: Firms state the call; class challenges it (13 min)

Resources: *Textbook, handouts, visuals, laptops.*

Google Classroom, projector, engineering notebook, Design Brief packet / worksheets, laptops or phones for AI tools.

Re-Teaching Activities: *What you prepare if students do not master the objective.*

Review during bellwork; small-group re-teach; peer support.

Reflection: *Students analyze what they accomplished and how they reached mastery.*

Exit ticket – students state what they accomplished and what worked best.

Study Skills: *Skills implemented by students – note-taking, summarizing, teamwork, etc.*

Note-taking, summarizing, outlining, reading comprehension, teamwork, oral communication, problem-solving.

SPED & 504 Accommodations / Modifications: *Codes.*

Per Individualized Education Plan (IEP) / 504 plan.

ELL Accommodations: *Codes.*

Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP).